Solutions to Homework #8

Problem 1

(a) By the definition, we have

$$\begin{aligned}
Y_c(j\Omega, \tau) &= \int_{-\infty}^{\infty} y(t)w(t-\tau)e^{-j\Omega t}dt \\
&= \int_{-\infty}^{\infty} x(t-d)w(t-\tau)e^{-j\Omega t}dt \\
&\overset{(a)}{=} \int_{-\infty}^{\infty} x(s)w(s+d-\tau)e^{-j\Omega(s+d)}ds \\
&= e^{-j\Omega d}\int_{-\infty}^{\infty} x(s)w(s-(\tau-d))e^{-j\Omega s}ds \\
&= e^{-j\Omega d}X_c(j\Omega, \tau-d).
\end{aligned}$$

(b) Again, by the definition, we have

$$\begin{aligned}
Z_c(j\Omega, \tau) &= \int_{-\infty}^{\infty} z(t)w(t-\tau)e^{-j\Omega t}dt \\
&= \int_{-\infty}^{\infty} x(t)e^{j\Omega_0 t}w(t-\tau)e^{-j\Omega t}dt \\
&= \int_{-\infty}^{\infty} x(t)w(t-\tau)e^{-j(\Omega-\Omega_0)t}dt \\
&= X_c(j(\Omega-\Omega_0), \tau).
\end{aligned}$$

(c) By the definition of the Fourier transform, we have

$$\mathcal{F}\{x(t)w(t-\tau)\} = X_c(j\Omega, \tau).$$

By the conjugation property, it implies

$$\mathcal{F}\{x^*(t)w^*(t-\tau)\} = X_c^*(-j\Omega, \tau).$$

Then by the multiplication property we have

$$\int_{-\infty}^{\infty} |x(t)|^2|w(t-\tau)|^2 e^{-j\Omega t}dt = \frac{1}{2\pi}\int_{-\infty}^{\infty} X_c(j\nu, \tau)X_c^*(j(\nu-\Omega), \tau)d\nu.$$

Particularly, when $\Omega = 0$, it reduces to

$$\begin{aligned}
\int_{-\infty}^{\infty} |x(t)|^2|w(t-\tau)|^2 dt &= \frac{1}{2\pi}\int_{-\infty}^{\infty} X_c(j\nu, \tau)X_c^*(j(\nu), \tau)d\nu \\
&= \frac{1}{2\pi}\int_{-\infty}^{\infty} |X_c(j\nu, \tau)|^2 d\nu \\
&= \frac{1}{2\pi}\int_{-\infty}^{\infty} |X_c(j\nu, \Omega)|^2 d\Omega.
\end{aligned}$$

By taking the integral with respect to τ on both sides, we obtain

$$\int_{-\infty}^{\infty}\int_{-\infty}^{\infty}|x(t)|^2|w(t-\tau)|^2dtd\tau = \frac{1}{2\pi}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty}|X_c(j\nu,\Omega)|^2d\Omega d\tau.$$

We do change of variable with $s = t-\tau$ for fixed t. Then the left-hand side becomes

$$\int_{-\infty}^{\infty}\int_{-\infty}^{\infty}|x(t)|^2|w(s)|^2dtds = \underbrace{\int_{-\infty}^{\infty}|w(s)|^2ds}_{1}\cdot\int_{-\infty}^{\infty}|x(t)|^2dt.$$

Problem 2

(a) Fix l and let $z[n] = x[n]w[n-lM]$. Then $X[k,l]$ is obtained by sampling the DTFT of $z[n]$ by

$$X[k,l] = \sum_{n=-\infty}^{\infty} z[n]e^{-j\frac{2\pi}{N}kn} = Z(e^{j\omega})|_{\omega=\frac{2\pi}{N}k}.$$

Therefore, the inverse DFT written as

$$\begin{aligned}\frac{1}{N}\sum_{k=0}^{N-1}X[k,l]e^{j\frac{2\pi}{N}kn} &= \frac{1}{2\pi}\int_{-\pi}^{\pi}Z(e^{j\omega})\sum_{m=-\infty}^{\infty}\delta\left(\omega-\frac{2\pi m}{N}\right)e^{-j\omega n}d\omega\\ &= z[n]*\sum_{m=-\infty}^{\infty}\delta[n-Nm].\end{aligned}$$

Note that by the construction, $z[n]$ is supported on the interval $[lM, lM+L-1]$ of length L. The convolution with the discrete-time pulse train will add aliased copies given by shifts in multiples of N. To avoid overlaps, we need to satisfy $N \geq L$. In other words, the size of DFT should be no less than the window length. If this is the case, the inverse DFT and $z[n]$ coincide on the support of the window. Therefore we have

$$z[n]w[n-lM] = x[n]w^2[n-lM] = \left(\frac{1}{N}\sum_{k=0}^{N-1}X[k,l]e^{j\frac{2\pi}{N}kn}\right)w[n-lM].$$

(b) From part (a), we have

$$x[n]\sum_{l=-\infty}^{\infty}w^2[n-lM] = \frac{1}{N}\sum_{l=-\infty}^{\infty}\sum_{k=0}^{N-1}X[k,l]w[n-lM]e^{j\frac{2\pi}{N}kn}.$$

It remains to show that

$$\sum_{l=-\infty}^{\infty}w^2[n-lM] = 1.$$

Since $L = 2M$, by the construction, the left-hand side corresponds to a M-periodic signal. Therefore, it suffices to show the identity for n in the interval $[0, M-1]$. For those n, the only active summands are $w[n]$ and $w[n+M]$. A sufficient condition is given as

$$w^2[n] + w^2[n+M] = 1, \quad n = 0, 1, \dots, M-1.$$

(c) If $w[n] = \frac{1}{\sqrt{2}}$ for all $n \in [0, L-1]$, then it satisfies the condition in part (b).

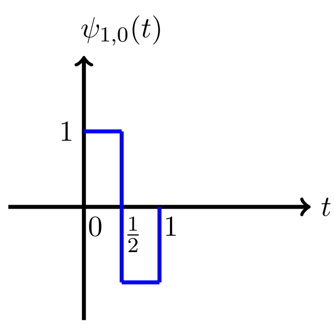
(d) Let

$$w[n] = \sqrt{\frac{M - 0.5 - |n - M + 0.5|}{M-1}}, \quad n = 0, 1, \dots, 2M-1.$$

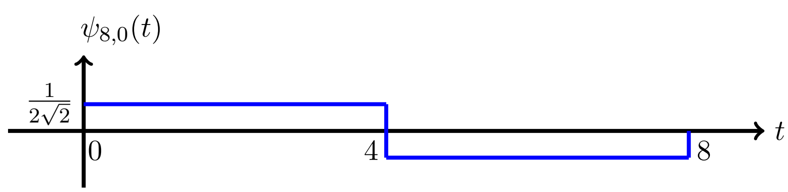
Then it also satisfies the condition in part (b).

Problem 3

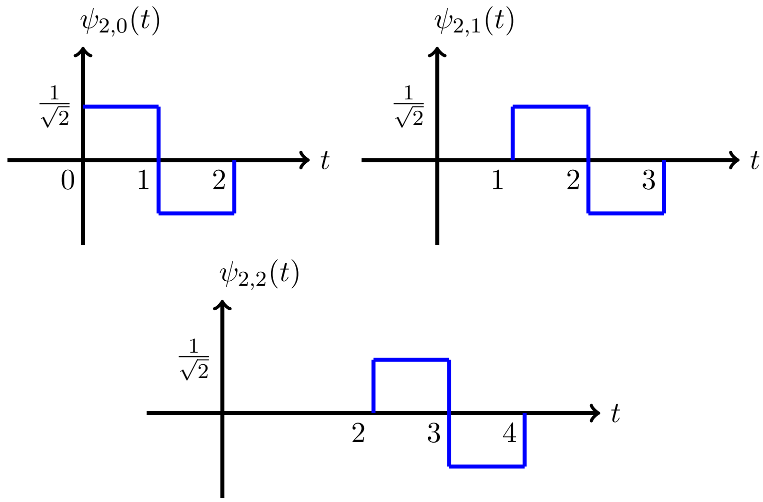
(a) The graph of $\psi_{1,0}(t)$ is plotted as follows.



(b) The graph of $\psi_{8,0}(t)$ is plotted as follows.



(c) The graph of $\psi_{2,0}(t)$, $\psi_{2,1}(t)$, and $\psi_{2,2}(t)$ are plotted as follows.



(d) The product of $\psi_{2,0}(t)$ and $\psi_{8,0}(t)$ satisfies

$$\psi_{2,0}(t)\psi_{8,0}(t) = \psi_{2,0}(t).$$

Therefore, they satisfy

$$\int_{-\infty}^{\infty} \psi_{2,0}(t)\psi_{8,0}(t)dt = 0.$$

Similarly, we also have

$$\int_{-\infty}^{\infty} \psi_{2,2}(t)\psi_{8,0}(t)dt = 0.$$

Finally, since

$$\psi_{2,0}(t)\psi_{2,2}(t) = 0,$$

it follows that

$$\int_{-\infty}^{\infty} \psi_{2,0}(t)\psi_{2,2}(t)dt = 0.$$

We have shown that the three signals are pairwise orthogonal.

Problem 4

(a) We apply the Cauchy-Schwarz inequality for $g(t) = tx(t)$ and $f(t) = \frac{dx(t)}{dt}$ so that

$$\left| \int_{-\infty}^{\infty} (tx(t))^* \frac{dx(t)}{dt} dt \right|^2 \leq \int_{-\infty}^{\infty} |tx(t)|^2 \, dt \int_{-\infty}^{\infty} \left| \frac{dx(t)}{dt} \right|^2 dt.$$

Furthermore, since t and $x(t)$ are real-valued, $(tx(t))^* = tx(t)$. Therefore, we obtain

$$\left| \int_{-\infty}^{\infty} tx(t) \frac{dx(t)}{dt} dt \right|^2 \leq \int_{-\infty}^{\infty} t^2 x^2(t) dt \int_{-\infty}^{\infty} \left(\frac{dx(t)}{dt} \right)^2 dt.$$

(b) Recall that E is defined by

$$E = \int_{-\infty}^{\infty} |b(t)|^2 \, dt.$$

We apply the change of variable by $t = z + \mu_t$, where μ_t is a fixed constant. Then it follows that $dt = dz$. Since $b(z + \mu_t) = x(z)$, we obtain

$$E = \int_{-\infty}^{\infty} |b(z + \mu_t)|^2 \, dz = \int_{-\infty}^{\infty} |x(z)|^2 \, dz = \int_{-\infty}^{\infty} |x(t)|^2 \, dt.$$

Since $b(t)$ is real valued, $B_c(j\Omega)$ has conjugate symmetry, which implies

$$|B_c(-j\Omega)| = |B_c(j\Omega)|, \quad \forall \Omega.$$

Therefore, we have

$$\begin{aligned}
\mu_\Omega &= \frac{1}{2\pi E} \int_{-\infty}^{\infty} \Omega \, |B_c(j\Omega)|^2 \, d\Omega \\
&= \frac{1}{2\pi E} \left(\int_{-\infty}^{0} \Omega \, |B_c(j\Omega)|^2 \, d\Omega + \int_{0}^{\infty} \Omega \, |B_c(j\Omega)|^2 \, d\Omega \right) \\
&= \frac{1}{2\pi E} \left(\int_{-\infty}^{0} \Omega \, |B_c(j\Omega)|^2 \, d\Omega - \int_{-\infty}^{0} \Omega \, |B_c(j\Omega)|^2 \, d\Omega \right) = 0.
\end{aligned}$$

(c) Let $z = t + \mu_t$. Then we have $dz = dt$. By this change of variables, we obtain

$$\int_{-\infty}^{\infty} t^2 x^2(t) dt = \int_{-\infty}^{\infty} t^2 b^2(t + \mu_t) dt = \int_{-\infty}^{\infty} (z - \mu_t)^2 b^2(z) dz = E\sigma_t^2.$$

Using the fact that $j\Omega X_c(j\Omega)$ is the Fourier transform of $dx(t)/dt$ and Parseval's identity, we have

$$\begin{aligned}
\int_{-\infty}^{\infty} \left(\frac{dx(t)}{dt} \right)^2 dt &= 2\pi \int_{-\infty}^{\infty} |j\Omega X_c(j\Omega)|^2 \, d\Omega \\
&= 2\pi \int_{-\infty}^{\infty} \Omega^2 \, |B_c(j(\Omega + \mu_\Omega)|^2 \, d\Omega \\
&= 2\pi \int_{-\infty}^{\infty} (\Lambda - \mu_\Omega)^2 \, |B_c(j(\Omega))|^2 \, d\Lambda \qquad \Lambda = \Omega + \mu_\Omega, \ d\Lambda = d\Omega \\
&= 2\pi \int_{-\infty}^{\infty} (\Omega - \mu_\Omega)^2 \, |B_c(j(\Omega))|^2 \, d\Omega \qquad \Omega = \Lambda, d\Omega = d\Lambda \\
&= E\sigma_\Omega^2,
\end{aligned}$$

where the second identity holds since $\mu_\Omega = 0$ and $|X_c(j\Omega)| = |B_c(j\Omega)|$.

(d) By the integration by parts with $u(t) = tx(t)$, $\frac{du(t)}{dt} = t\frac{dx(t)}{dt} + x(t)$, $v(t) = x(t)$, and $\frac{dv(t)}{dt} = \frac{dx(t)}{dt}$, we obtain

$$\begin{aligned}\int_{-\infty}^{\infty} tx(t)\frac{dx(t)}{dt}dt &= tx^2(t)|_{-\infty}^{\infty} - \int_{-\infty}^{\infty} x(t)\left(t\frac{dx(t)}{dt} + x(t)\right)dt \\ &= 0 - \int_{-\infty}^{\infty} tx(t)\frac{dx(t)}{dt}dt - \int_{-\infty}^{\infty} x^2(t)dt.\end{aligned}$$

By rearranging the summands, we have

$$2\int_{-\infty}^{\infty} tx(t)\frac{dx(t)}{dt}dt = -\int_{-\infty}^{\infty} x^2(t)dt = -E.$$

Therefore, we conclude that

$$\int_{-\infty}^{\infty} tx(t)\frac{dx(t)}{dt}dt = -\frac{E}{2}.$$

(e) Plugging in the previous parts into the inequality in Part (a) provides

$$\left|\frac{1}{2}E\right|^2 \leq E^2\sigma_t^2\sigma_\Omega^2,$$

which simplifies to

$$\frac{1}{2} \leq \sigma_t\sigma_\Omega.$$

In applying the Cauchy-Schwarz inequality, we let $g(t) = tx(t)$ and $f(t) = \frac{dx(t)}{dt}$. Since the equality holds if and only if $g(t) = cf(t)$ for some constant c. The condition is rewritten as

$$tx(t) = c\frac{dx(t)}{dt},$$

which is equivalent to

$$\frac{\frac{dx(t)}{dt}}{x(t)} = \frac{t}{c}.$$

By taking the anti-derivative, we obtain

$$\log x(t) = \frac{t^2}{c} + D$$

for some constant D. Therefore, $x(t)$ is written as

$$x(t) = e^{\frac{t^2}{c}+D} = e^D e^{\frac{1}{c}t^2}.$$

We obtain the desired form by letting $A = e^D$ and $-\alpha = \frac{1}{c}$. The condition $\alpha > 0$ is required to satisfy that $\lim_{t\to\infty} tb(t) = 0$.

(f) By plugging in the expression of $b(t)$, we obtain

$$\begin{aligned}\mu_t &= 2\pi \int_{-\infty}^{\infty} \frac{1}{2\pi} \cdot t|w(t-\tau_0)|^2 dt \\ &= \int_{-\infty}^{\infty} (s+\tau_0)|w(s)|^2 ds \\ &= \tau_0 \underbrace{\int_{-\infty}^{\infty} |w(s)|^2 ds}_{1} + \underbrace{\int_{-\infty}^{\infty} s|w(s)|^2 ds}_{0}.\end{aligned}$$

Here the second summand is 0 because $s|w(s)|^2$ is an odd function. Next,

$$\sigma_t^2 = 2\pi \int_{-\infty}^{\infty} \frac{(t-\tau_0)^2|w(t-\tau_0)|^2}{2\pi} dt = \int_{-\infty}^{\infty} s^2|w(s)|^2 ds = \int_{-\infty}^{\infty} t^2|w(t)|^2 dt.$$

Next, note that

$$B_c(j\Omega) = \frac{1}{\sqrt{2\pi}} W_c(j(\Omega-\Omega_0))e^{-j\tau_0(\Omega-\Omega_0)}.$$

Therefore,

$$|B_c(j\Omega)| = \frac{1}{\sqrt{2\pi}} |W_c(j(\Omega-\Omega_0))|.$$

By plugging in this to the definition, we have

$$\mu_\Omega = \frac{2\pi}{2\pi} \int_{-\infty}^{\infty} \frac{1}{2\pi} \cdot \Omega |W_c(j(\Omega-\Omega_0))|^2 d\Omega = \int_{-\infty}^{\infty} \frac{1}{2\pi} \cdot \Omega |W_c(j(\Omega-\Omega_0))|^2 d\Omega.$$

By the change of variable of $\nu = \Omega - \Omega_0$, we continue as

$$\begin{aligned}\mu_\Omega &= \int_{-\infty}^{\infty} \frac{1}{2\pi} \cdot (\nu+\Omega_0)|W_c(j(\nu))|^2 d\nu \\ &= \Omega_0 \cdot \underbrace{\frac{1}{2\pi} \int_{-\infty}^{\infty} |W_c(j(\nu))|^2 d\nu}_{1} + \underbrace{\frac{1}{2\pi} \int_{-\infty}^{\infty} \nu |W_c(j(\nu))|^2 d\nu}_{0},\end{aligned}$$

where the last identity holds since $\nu|W_c(j(\nu))|^2$ is an odd function and Parseval's theorem implies

$$\frac{1}{2\pi} \int_{-\infty}^{\infty} |W_c(j(\Omega))|^2 d\Omega = \int_{-\infty}^{\infty} |w(t)|^2 dt = 1.$$

(g) We are given $w(t) = \sqrt{3/(2c)}\mathrm{tri}(t/c)$. Note the

$$w(t) = \sqrt{\frac{3}{2c}} \cdot \left(1 - \frac{t}{c}\right), \quad t \geq 0$$

and $w(-t) = w(t)$, that is, $w(t)$ is an even function.

First we compute σ_t by

$$\begin{aligned}
\sigma_t^2 &= \int_{-\infty}^{\infty} t^2 |w(t)|^2 dt \\
&\overset{(c)}{=} 2\int_{0}^{\infty} t^2 |w(t)|^2 dt \\
&= \int_0^c t^2 \cdot \frac{3}{c} \cdot \left(1 - \frac{t}{c}\right)^2 dt \\
&= \frac{3}{c^3} \int_0^c (t^4 - 2ct^3 + c^2t^2) dt \\
&= \frac{3}{c^3} \cdot \left(\frac{c^5}{5} - \frac{c \cdot c^4}{2} + \frac{c^2 \cdot c^3}{3}\right) \\
&= \frac{3}{c^3} \cdot \frac{c^5}{30} = \frac{c^2}{10}.
\end{aligned}$$

Next, σ_Ω is computed by

$$\begin{aligned}
\sigma_\Omega^2 &= \frac{1}{2\pi} \int_{-\infty}^{\infty} \Omega^2 |W_c(j\Omega)|^2 d\Omega \\
&= \frac{1}{2\pi} \int_{-\infty}^{\infty} |j\Omega W_c(j\Omega)|^2 d\Omega \\
&= \int_{-\infty}^{\infty} \left| \frac{d}{dt} w(t) \right|^2 dt \\
&= \int_{-c}^{0} \left| \sqrt{\frac{3}{2c}} \cdot \frac{1}{c} \right|^2 dt + \int_0^c \left| \sqrt{\frac{3}{2c}} \cdot \left(-\frac{1}{c}\right) \right|^2 dt \\
&= 2\int_0^c \frac{3}{2c^3} dt \\
&= 2 \cdot \frac{3}{2c^2} = \frac{3}{c^2}.
\end{aligned}$$

Therefore

$$\sigma_t \sigma_\Omega = \sqrt{\frac{c^2}{10} \cdot \frac{3}{c^2}} = \sqrt{\frac{3}{10}} > \frac{1}{2}.$$

This implies that the time-frequency resolution trade-off by a triangular window is sub-optimal.